

1(a)	1 mark for each correct row ; ; ; ;	4																											
	<table><tr><th rowspan="2">statement</th><th colspan="3">process</th></tr><tr><th>active transport</th><th>facilitated diffusion</th><th>simple diffusion</th></tr><tr><td>movement of oxygen into a red blood cell</td><td>x</td><td>x</td><td>✓</td></tr><tr><td>occurs in both animal and plant cells</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>uses carrier proteins</td><td>✓</td><td>✓</td><td>x</td></tr><tr><td>movement of non-polar molecules between the fatty acid tails of the phospholipid molecules</td><td>x</td><td>x</td><td>✓</td></tr><tr><td>movement of ions down a concentration gradient</td><td>x</td><td>✓</td><td>x</td></tr></table>	statement	process			active transport	facilitated diffusion	simple diffusion	movement of oxygen into a red blood cell	x	x	✓	occurs in both animal and plant cells	✓	✓	✓	uses carrier proteins	✓	✓	x	movement of non-polar molecules between the fatty acid tails of the phospholipid molecules	x	x	✓	movement of ions down a concentration gradient	x	✓	x	
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1(b)(i)	a line drawn to any of the 4 black rectangles shown in the cell wall of the cells three from the right and labelled with the letter C ;	1																											
1(b)(ii)	provides, large / high, surface area or provides a, large / high, surface area : volume (ratio) ;	1																											
1(b)(iii)	apoplastic / apoplast ; I cell wall route	1																											

Question	Answer	Marks
1(c)	any three from: 1 cohesion / hydrogen bonding, between water molecules ; 2 detail about H bonding ; e.g. oxygen has, small / δ negative charge, hydrogen has small / δ positive charge water molecules are polar so attract each other 3 adhesion / hydrogen bonding, between water molecules and, cell wall / xylem wall / cellulose / hydrophilic regions of lignin ; 4 adhesion / AW), supports the column of water ; 5 transpiration / evaporation (from leaves), pulls the column or ref. to tension from, above / leaves or ref. to water potential gradient from roots to leaves ;	3